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$$\vec{\tau}_1 + \vec{\tau}_2 + \vec{\tau}_3 + \cdots + \vec{\tau}_n = 0 \qquad (5.31)$$

Thus, we can also conclude that a rigid body is in mechanical equilibrium when the net force and net torque acts on the body is zero.

$$\vec{F}_{net} = 0 \text{ and } \vec{\tau}_{net} = 0 \qquad (5.32)$$

As the forces and torques are vector quantities, the directions are to be taken with proper sign conventions.

5.3.1 Types of Equilibrium

Based on the above discussions, we come to a conclusion that different types of equilibrium are possible based on the different conditions. They are consolidated in Table 5.2.

Table 5.2 Different types of Equilibrium and their Conditions.

Type of equilibrium	Conditions
Translational equilibrium	▪ Linear momentum is constant. ▪ Net force is zero.
Rotational equilibrium	▪ Angular momentum is constant. ▪ Net torque is zero.
Static equilibrium	▪ Linear momentum and angular momentum are zero. ▪ Net force and net torque are zero.
Dynamic equilibrium	▪ Linear momentum and angular momentum are constant. ▪ Net force and net torque are zero.
Stable equilibrium	▪ Linear momentum and angular momentum are zero. ▪ The body tries to come back to equilibrium if slightly disturbed and released. ▪ The center of mass of the body shifts slightly higher if disturbed from equilibrium. ▪ Potential energy of the body is minimum and it increases if disturbed.
Unstable equilibrium	▪ Linear momentum and angular momentum are zero. ▪ The body cannot come back to equilibrium if slightly disturbed and released. ▪ The center of mass of the body shifts slightly lower if disturbed from equilibrium. ▪ Potential energy of the body is not minimum and it decreases if disturbed.
Neutral equilibrium	▪ Linear momentum and angular momentum are zero. ▪ The body remains at the same equilibrium if slightly disturbed and released. ▪ The center of mass of the body does not shift higher or lower if disturbed from equilibrium. ▪ Potential energy remains same even if disturbed.